WHITE PAPER · YCM TV-188 B × AEROSPACE

YCM TV-188 B for Aerospace

Large-envelope CNC vertical mill for oversized workpieces, complex pocketing, and multi-face machining. This paper covers what the machine does, the aerospace parts we produce with it, the alloys we run, the tolerances and finishes aerospace buyers expect, and the documentation package.

32" W × 70" L × 32" H workpiece envelope	ENVELOPE
±0.0005" routine on critical features	TOLERANCE
1994	SINCE
Same-week / rig-down direct	RESPONSE

Loco Motive · 32" W × 70" L × 32" H workpiece envelope · large-envelope CNC vertical machining center

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Executive summary

Machine: YCM TV-188 B (shop nickname: *Loco Motive*). large-envelope CNC vertical machining center at B&R Productions, running from our New Waverly, Texas shop.

Application: Large-envelope CNC vertical mill for oversized workpieces, complex pocketing, and multi-face machining for Aerospace customers — Aerospace primes, tier-1 and tier-2 suppliers, MRO shops, propulsion component manufacturers, structural assembly integrators, actuator OEMs.

Envelope: 32" W × 70" L × 32" H workpiece envelope. **Tolerance:** ±0.0005" routine on critical features. **Traceability:** Material by heat and lot on every job.

The machine, in context

Loco Motive is where the oversized work lives — valve bodies, wellhead spool adapters, large manifold blocks, and any part that busts the 20-inch class of mills. YCM's TV-188 platform is a heavier, more rigid class of vertical machining center; the extra column and table mass matters when you're taking heavy roughing cuts in Inconel or hogging out a wellhead block.

Why Loco Motive for Aerospace work

Aerospace structural work is prismatic, often large, and frequently machined from solid with high material removal ratios. A 32" × 70" × 32" envelope with 25 HP at BT50 and a 4,409 lb table takes the structural parts that a 20-inch-class mill can't, and does it with the rigidity that heavy titanium and Inconel removal demands.

What this looks like on a real part

A large structural fitting machined from a solid titanium block is mostly an exercise in removing material without putting heat into what's left. Titanium's low thermal conductivity keeps the heat in the cut zone and the chip, which is fine — until the chip stops leaving. On a deep pocket at this size, evacuation is the whole game: a chip that sits and re-cuts will both destroy finish and dump heat back into a part you have been carefully keeping cool. The T-base helps, the toolpath has to do the rest.

> **When this is the wrong machine**
>
> Not a high-RPM machine. Thin-wall aluminum aerospace work that wants 12,000+ RPM and light, fast passes isn't what this platform is for. Smaller brackets and fittings run more economically on Pistol Whiskers.

Full specifications

Model	YCM TV-188 B ("Loco Motive")
Type	3-axis CNC vertical machining center — heavy-duty class

Workpiece envelope	32" W × 70" L × 32" H (B&R working spec)
Axis travels	70.9" X × 33.9" Y × 29.5" Z
Table size	78.7" × 33.9" (2,000 × 860 mm)
Max workpiece weight	4,409 lb (2,000 kg)
Spindle taper	BT50
Spindle speed	6,000 RPM standard (10,000 RPM optional)
Spindle power	25 HP standard (30 HP optional), 30-min rating
Automatic tool changer	24 tools standard (40 optional)
Rapid traverse	591 IPM; cutting feed 197 IPM
Control	Fanuc MXP-200i
Structure	Patented T-base for rigidity + chip disposal
Tolerance capability	±0.0005" routine on critical features
Materials	Stainless, carbon steel, tool steel, Inconel, Hastelloy, aluminum, titanium

Values marked "platform" reflect the machine manufacturer's published spec range for this platform family; B&R's working spec is what we quote on this specific unit.

Who this serves in Aerospace

Typical buyer: Aerospace primes, tier-1 and tier-2 suppliers, MRO shops, propulsion component manufacturers, structural assembly integrators, actuator OEMs.

What's hard about aerospace work: Certification-driven documentation, exotic superalloys, tight tolerances on structural and engine components, and finding a job shop that understands aged-condition superalloys rather than learning on the part.

Typical Aerospace parts we produce on Loco Motive

- Large structural brackets and airframe fittings
- Manifold blocks and valve housings for hydraulic systems
- Actuator housings and pylon components
- Engine mount hardware requiring extended envelope
- Large tooling and fixtures for aerospace assembly

Above list is representative; if your part isn't shown, call and ask — most oilfield / aerospace / defense / industrial turned or milled work fits somewhere in the shop.

Alloys we run for Aerospace

Standard range: Titanium Grade 5 (Ti-6Al-4V) and Grade 23 (Ti-6Al-4V ELI) · Inconel 718 · Inconel 625 · Waspaloy · A286 · 17-4 PH · 15-5 PH · 13-8 Mo PH · Aluminum 7075/6061 · Custom aerospace alloys on request.

Titanium Grade 5 is the workhorse and it's temperature-sensitive: aggressive chip load with flood coolant keeps the heat in the chip, not the workpiece. Light cuts overheat titanium and cook tools; that's why titanium demands a different intuition than steel. Inconel 718 in aerospace-aged condition is harder and less forgiving than the annealed billet stock — most aerospace 718 work assumes aged. 17-4 PH and 15-5 PH are common for structural and actuator components; both are precipitation-hardening stainless with condition-specific machining plans.

> **What "we run these weekly" means:** we stock or reliably source the common alloys in the industry's typical spec range. Sharp tooling, proven feeds and speeds, and process control specific to each alloy family. Your part is not the one we're experimenting on.

How we machine it — our 5-step process

This is the process B&R runs on every job, aerospace or otherwise. The specifics of feeds, speeds, and tooling shift by alloy and machine; the discipline is universal.

1. **Material verification.** Every job starts with material traceability. Heat and lot numbers documented against the mill test report. For customer-supplied material, we verify against the CofC before touching the stock.

2 **First-article layout.** Stock is measured and laid out before any cutting starts. Concentricity, roundness, and squareness of the starting material established; if the stock is out of tolerance we call the customer before wasting time on a bad blank.

3 **Roughing with process control for the alloy.** Feeds, speeds, and coolant strategy set for the specific alloy family — sharp coated carbide (or PCD for high-volume Ti Grade 5), positive rake, high-pressure through-tool coolant on 2507 and Inconel to control cutting temperature. Interrupted cuts get extra attention because they cycle heat into the workpiece.

4 **Finish pass to spec.** Finish pass with a fresh insert, controlled DOC, and measured surface-finish targets. Critical features (bore concentricity, seat grooves, sealing surfaces) are held tighter than most job-shop lathes can offer.

5 **CMM verification + documentation.** Every critical dimension CMM-verified. First-article report generated. Certificate of conformance issued with delivery. If your quality group needs a documentation package beyond that, tell us with the RFQ and we'll be straight about what we can and can't produce in-house.

Tolerance and finish expectations in Aerospace

±0.0005" routine on critical features. Tighter with proper fixture and setup — some aerospace features run ±0.0002" or GD&T-driven cylindricity and position tolerances. CMM verification standard on first articles; process-driven measurement on runs. Surface finish measured, not estimated.

Documentation and quality workflow

First-article layout and CMM verification standard. Material traceability by heat and lot. Certificates of conformance issued with delivery. Proprietary prints handled under NDA — prints stay in-shop and aren't sent to outside vendors. To be clear about what we are not: B&R is not AS9100-registered and not ITAR-registered. If your program requires either, you need a registered shop and we'll say so on the first call rather than after you've placed the order.

Response time and emergency work

AOG (aircraft-on-ground) work prioritized when material is available. Emergency response supported for existing aerospace customers with blanket arrangements or established relationships. Call direct at (936) 291-7827.

How we compare

Compared to a captive aerospace shop: we're faster for one-offs and small runs, more flexible on setup, and we don't require a long approval process. Compared to a lowest-bidder commodity shop: we understand the alloys and we treat first-article inspection as a discipline rather than a formality. Where a registered shop beats us is on programs that require AS9100 flow-down — we're the wrong call for those, and the right call for non-controlled work, prototypes, tooling, and MRO parts where alloy competence and turnaround matter more than a certificate on the wall.

Working with B&R Productions

New customer or existing, the process starts the same way: call (936) 291-7827 or send a print through the quote form. Prints are accepted as PDF, STEP, IGES, DWG, or DXF; if you only have a sample part, we reverse-engineer routinely. NDAs on request, and prints stay in-shop rather than going out to third-party vendors. Note for defense and aerospace buyers: B&R is not ITAR- or AS9100-registered, so export-controlled prints and AS9100 flow-down programs need a registered shop — we'll tell you that on the first call, not after the PO.

Quotes typically come back within 24 hours for standard work, same-day for repeat customers on stocked materials, and near-immediately by phone for emergency and rig-down situations. Lead times are quoted honestly — we don't promise Wednesday and deliver next month.

Frequently asked questions

8 questions covering the YCM TV-188 B platform and Aerospace work. Also indexed as FAQ schema for AI answer engines.

What's the max part size on the YCM TV-188B?

B&R's stated working envelope is 32" W × 70" L × 32" H. The YCM TV-188B platform has a 78.7" × 33.9" table, travels of 70.9" X × 33.9" Y × 29.5" Z, and max workpiece weight of 4,409 lb (2,000 kg).

Does the TV-188B have a high-speed spindle option?

Standard spindle is 6,000 RPM with a 10,000 RPM option on some builds. BT50 tooling, 25 HP standard (30 HP optional, 30-min rating).

What alloys can the TV-188B run at production speeds?

Full oilfield + aerospace alloy range including Inconel and Hastelloy. The T-base structural design and heavy castings keep it stable through the heavy roughing cuts these alloys demand.

Do you serve aerospace primes and tier-1/2 suppliers?

On work that doesn't require a registered QMS, yes — that's the honest boundary. B&R is not AS9100-registered, and where that matters it matters absolutely, so we say it before you spend time on an RFQ: if your program requires AS9100 flow-down, we're the wrong shop. Prototypes, tooling, fixtures, MRO and repair parts, and non-controlled components are where a shop like this is genuinely useful to a prime or a tier-1, and that's the work we'd like to quote.

Can you handle ITAR-controlled aerospace/defense work?

No. ITAR compliance requires registration with the U.S. State Department's DDTC, and B&R does not hold that registration. If your print is ITAR-controlled, you need a registered shop — send it elsewhere and we'd rather tell you now than after you've placed an order. Plenty of aerospace and defense work isn't export-controlled, and that work we quote gladly. Either way, proprietary prints are handled under NDA and stay in-shop.

What tolerances do you hold on aerospace parts?

±0.0005" routine on critical features. First-article layout and CMM verification standard. Surface-finish targets measured to spec.

What aerospace alloys do you run?

Titanium Grade 5 (Ti-6Al-4V), Grade 23 ELI, Inconel 718 and 625, Waspaloy, A286, 17-4 PH, 15-5 PH, aluminum 7075/6061. Custom aerospace alloys on request.

Can you respond to AOG (aircraft-on-ground) work?

Yes, when material is available. Emergency response supported for existing aerospace customers. Newer customers should call to discuss what's realistic for their specific alloy and part.

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Ready to talk about your Aerospace work?

Send us a print or a sample photo and specs. Emergencies: call direct and say "rig-down."

Published by B&R Productions — a precision CNC machining shop in New Waverly, Texas, in business since 1994. Serving oil & gas, aerospace, defense, and industrial customers across Texas and the Gulf Coast.

Written by the B&R Productions team. Machine specs verified against manufacturer data. Alloy and process notes reflect shop practice as of 2026-01-15.